

Education

University of Pennsylvania: 2026-2028 | M.S.E. in **Scientific Computing**

Virginia Tech: 2023 - 2026 | B.S. in **Computational Modeling and Data Analytics** Minor in **Mathematics, Computer Science** | GPA: 3.85

Relevant Courses: Statistical Methods | Probability and Distributions | Linear Algebra | Software Design and Data Structures | Computer Organization | Data Structures and Algorithms | Machine Learning | Mathematical Modeling I, II | Data Analytics and Visualization | Parallel Computing | Intro to Programming | Advanced Programming | Discrete Structures

Research Experience

Technical University of Munich

Computational Geophysics Visiting Researcher | 07/2025 - 08/2025

- Developed new coupled Tsunami Landslide simulations using Peano4 using Python and C++ utilizing finite volume and ADER-DG PDE solvers. Presented this work to TUM leadership, paving the way for real world tsunami-landslide comparisons.
- Used ExaHype2 specific limiter solutions to smooth oscillations within the grid solutions for the coupled simulations, creating quad solver simulations by modularizing solver focus.
- Funded by the *Steeneck Experiential Learning Scholarship* through Virginia Tech.

Virginia Tech

Inverse Problem Undergraduate Researcher | 01/2025 - 06/2026

- Research into solving physical modeling inverse problems for the FitzHugh-Nagumo ODE and the Hodgkins Huxley model using different neural network architectures in PyTorch, allowing for efficient multi-parameter prediction with a 97%+ accuracy.

Machine Learning Dimensionality Reduction Undergraduate Researcher | 08/2024 - 06/2026

- Developed autoencoder based methods for ODE input data to create a lower dimensionality representation for inverse problem mapping focused neural networks, reducing the computational cost of training the inference neural networks.
- Achieved a 99%+ accuracy on parameter predictions with autoencoder lower dimensionality data.
- *Awarded \$500 through the CMDA 2024 Fall Undergraduate Research Grant for continuation of the research.*
- Presented the work in a talk format at *New Connections in Mathematics 2024 at Duke University (funded by NSF to attend)* as well as *SUMS 2024 at James Madison University*
- Presented the work in a talk format at *Mid Atlantic Regional Math Alliance conference at UVA* as well as *Mathematical Association of America MD-DC-VA Spring Conference at George Mason University*
- Competed in the *Layman Prize Competition at Virginia Tech*, presenting the work on autoencoders against five other undergraduates involved in math research

Code Generation Toolkit Undergraduate Researcher | 09/2023 - 07/2025

- Using tree graph methodology to create a code generation platform using Python to minimize the variants resulting from hardware computation algorithmic variation.
- Led to a 72% code reduction from implementation into the *Flash-X Physics Engine* in collaboration with Argonne National Lab.
- Awarded \$835 through the *CMDA 2024 Spring Undergraduate Research Grant* for the Code Generation research.
- Attended the Mathematical Association of America Spring 2024 meeting presenting the work of *CGKit: Code Generation Toolkit for Performant and Maintainable Variants of Source Code* resulting in a second place for student posters.

James Madison University

Jet Breakaway and Fluid Stream Modeling Undergraduate Researcher | 02/2023 - 05/2023

- Utilized data analytic and linear programming skills through MATLAB to research and model a fluid stream on a tulip flare tip related to exhaust fumes, culminating in a conference paper presented at the Mathematical Association of America conference.
- Investigated the relationship between the tulip flare tip width and pressure on jet breakaway shock shells to aid rocket launch gas and fluid stream dispersion technology. This research allows further analysis to be conducted into how to decrease noise production from rocket launches as a way to prevent damage to surrounding areas.

Professional Experience

Marriott International

Associate Software Engineer | 08/2024 - 06/2026

- Driving the development of enhancements to the provisioning automation microservice.
- Deployed the UI using Harness to be available for all Marriott internal developers

Core Platform Software Engineering Intern | 06/2024 - 08/2024

- Developed a microservice using SpringBoot WebFlux to automate the provisioning of API's in different development lifecycle environments to different clients and channels, as well as automating the initial provisioning of an API to any given development lifecycle environment.
- Reduced the current processes from 30+ actions for the provisioning of API's to 4 actions, saving upwards of thousands of hours per year for Marriott employees.

Papers

Johann Rudi, Youngjun Lee, [Aidan H. Chadha](#), Mohamed Wahib, Klaus Weide, Jared P. O'Neal, Anshu Dubey, [CG-Kit: Code Generation Toolkit for Performant and Maintainable Variants of Source Code Applied to Flash-X Hydrodynamics Simulations](#), Future Generation Computer Systems, Volume 160, 2024.

[Aidan H. Chadha](#), [Stroke Prediction Using Clinical and Social Features in Machine Learning](#), arXiv:2501.00048, 2024.

Matthew Caulfield, [Aidan H. Chadha](#), Caroline Lubert, Hasan Hamdan, [Predicting the Jet Boundary of a Turbulent Coanda Wall Jet Using Experimental Test Results](#), Mathematical Association of America MD-DC-VA Spring 2023 Conference.

Presentations

[Autoencoders Applied to Inverse Problem Solution Mapping with Neural Networks](#), Layman Prize Competition at Virginia Tech, May 8th, 2025

[Parameter Estimation with Dense, Convolutional, and Autoencoder Neural Networks Applied to the FitzHugh-Nagumo ODE](#), Mathematical Association of America MD-DC-VA Spring Conference, Apr 12, 2025

[Parameter Estimation with Dense, Convolutional, and Autoencoder Neural Networks Applied to the FitzHugh-Nagumo ODE](#), Mid-Atlantic Math Alliance Regional Conference, Mar 17, 2025

[Parameter Estimation with Dense, Convolutional, and Autoencoder Neural Networks Applied to the FitzHugh-Nagumo ODE](#), SUMS Conference, Nov 2, 2024

[Parameter Estimation with Dense, Convolutional, and Autoencoder Neural Networks Applied to the FitzHugh-Nagumo ODE](#), New Connections 2024 Conference, Nov 1, 2024

[CGKit: Code Generation Toolkit for Performant and Maintainable Variants of Source Code](#), Mathematical Association of America MD-DC-VA Spring Conference, April 30, 2024

[Predicting the Jet Boundary of a Turbulent Coanda Wall Jet Using Experimental Test Results](#), James Madison University Undergraduate Mathematics Colloquium, May, 2023

[Predicting the Jet Boundary of a Turbulent Coanda Wall Jet Using Experimental Test Results](#), Mathematical Association of America MD-DC-VA Spring Conference, April 29, 2023

Honors and Awards

CMDA Outstanding Senior for Excellence in Research Award | Spring 2026

- Awarded as a recognition of my contributions to research within the CMDA department.

Steenek Experience Learning Scholarship | Spring 2025

- Awarded \$3000 in scholarship money to fund a summer research program at the Technical University of Munich through the Virginia Tech College of Science.

Computational Modeling and Data Analytics Undergraduate Grant | Fall 2024

- Awarded \$500 in grant money from the Virginia Tech CMDA department for research into *Solving Physical Modeling Inverse Problems Using Neural Networks*

Mathematical Association of America | 2nd Place Award for Student Posters | Spring 2024 | [Link](#)

- Awarded second place for student posters for the presentation of *CGKit: Code Generation Toolkit for Performant and Maintainable Variants of Source Code*

Computational Modeling and Data Analytics Undergraduate Grant | Spring 2024

- Awarded \$835 in grant money from the Virginia Tech CMDA department for continued research into *Code Generation Toolkit for Performant and Maintainable Variants of Source Code*

Mathematical Association of America | 2nd Place Award for Student Presentations | Spring 2023 | [Link](#)

- Awarded second place for student presentations for the presentation of *Predicting the Jet Boundary of a Turbulent Coanda Wall Jet Using Experimental Test Results*